NATTARAT SAMARTKIT



Software Development | Software Testing | Data Analysis

Bangkok, Thailand | (+66) 62-176-4306 | nattarat.kwan@gmail.com
GitHub: github.com/kwank6704

PROFILE

Computer Engineering student at Chulalongkorn University with hands-on experience in software testing, software development, and business analysis. Seeking opportunities as a programmer, software tester, or data analyst.

EDUCATION

Chulalongkorn University *2024 – May 2028 (Expected)*
Bachelor of Engineering, Computer Engineering and Digital Technology (CEDT) | GPAX: 3.75 / 4.00
Relevant Coursework: Data Structures & Algorithms, Database Systems, Software Engineering, Software Development Practice I & II, Data Science & Data Engineering, Embedded Systems, AI & Machine Learning

Triam Udom Suksa School *2021 – 2023*
Science and Mathematics Program

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, SQL, HTML, CSS
Web Development: React, Node.js, Express.js, Django, Spring Boot, Tailwind CSS, Bootstrap, JavaFX
Databases: PostgreSQL, MySQL, MongoDB
Data Analysis & ML: NumPy, Pandas, Matplotlib, scikit-learn (clustering, PCA)
Software Testing: Manual and automated test case development and execution, defect tracking, API testing with Postman
Tools: Git, GitHub, Docker, Postman, Figma, Visual Studio Code, Eclipse
Automation: n8n, Zapier, Make.com, Microsoft Power Automate
Languages: Thai (Native), English (Proficient), Mandarin (Basic)

WORK EXPERIENCE

Software Tester (Summer Intern → Part-time) | Playtorium Solutions Public Company Limited *2026 (4 months)*
- Developed and executed manual and automated test cases to ensure system quality, and tracked software defects.
- Offered a part-time extension after the summer internship (4 months in total).

Business Analyst Trainee (Summer) | Ditto (Thailand) Public Company Limited *Summer 2025 (3 months)*
- Analyzed business requirements and designed process workflows to bridge communication between stakeholders and developers, while concurrently working a part-time developer role.

Developer Trainee (Part-time, Short-term Project) | TEAM Consulting Engineering and Management PCL *Summer 2025*
- Assisted in coding, debugging, and documenting software modules for a short-term internal engineering project.

RESEARCH

Lead Researcher | Auditing Algorithmic Equity in Large Language Models *2026*
- Built a 360-question dataset (90 prompts in Standard Central Thai, translated into Northern, Northeastern, and Southern Thai and verified by native speakers) across healthcare, finance, and legal domains.
- Developed a hybrid LLM-as-a-Judge framework (GPT-4o and Typhoon) to score responses on bias and information quality, validated with human review.
- Found that dialect inputs consistently received lower-quality, less detailed responses, especially in healthcare and legal topics. View paper

PROJECTS

Databank for Cocoa Supply Chain | Capstone Project, 4-member team (in progress) *Jul 2026 – Present*
- Extending a cocoa farm research platform with a LINE OA AI chatbot (Messaging API, LIFF) for conversational farmer data collection and modernizing the existing Flutter/Kotlin farmer app, using Scrum in 2-week sprints; Phase II adds a knowledge base and computer-vision disease detection.
- Role: Full-stack / QA and Knowledge Base Lead. Own the test strategy and UAT coordination, write API and end-to-end tests, set up the CI pipeline, and lead the web submission-history feature and knowledge base.

Campground Booking System | Full-stack web application *2025*
- Built with React, Tailwind CSS, Node.js, Express.js, and MongoDB; features user authentication, booking with availability checks, payment gateway integration, and reviews.
- Tested REST APIs with Postman and deployed frontend and backend on Vercel.

Hotel Booking System | Full-stack web application *2025*

- Built with React, Bootstrap, Django, and PostgreSQL following Scrum; features hotel search and filters, booking and cancellation, and an admin panel.
- Conducted business and system analysis to refine requirements; designed relational (PostgreSQL) and document-based (MongoDB) schemas with indexing.

Sleep Environment Score with AI-Assisted Environment System | Embedded Systems *2025*

- 5-member team IoT system using ESP32 sensor and gateway nodes (temperature, humidity, light, sound), ESP-NOW, Firebase, and Blynk, with an AI model scoring the sleep environment to control an aroma diffuser automatically.
- Role: UX/UI Designer and Hardware Engineer.

Iris Clustering Analysis | AI & Machine Learning coursework *2026*

- Applied K-Means and hierarchical clustering with scikit-learn, using standardization, elbow and silhouette analysis, and PCA; both methods identified k = 3 (silhouette score 0.46).

Vaccine Management System Backend | Spring Boot, MySQL

- Built scheduling and record APIs with role-based access control (Admin, User, Doctor); tested with Postman.

Other projects: Traffy Fondue Task Time Analysis (Data Science & Data Engineering, 2025); Bank Account Management System (Java, MySQL, JavaFX); CEDT Game (JavaFX); Fuellog Automation System (n8n); AI-Connected Web Interface (OpenAI API); OCR Web API (Typhoon OCR); Car Rental Front-End (React).

More projects: Project portfolio (Google Drive)

ACHIEVEMENTS & CERTIFICATE

- **Top 5, Women Thailand Cyber Top Talent 2025** *(2025)*
- **UX/UI Design** | BRAND'S Young Blood 2025 *(2025)*
- **Top 1% in A-Level Mathematics and Top 8% in A-Level English** *(2024)*
- **Google AI Essentials** | Coursera, issued by Google *(2024)*

LEADERSHIP & ACTIVITIES

- **Teacher Assistant** | Chulalongkorn University *(2025 – 2026)*
- **Teacher Assistant** | Seminar for Industrial Estate Authority of Thailand (IEAT) *(2025)*
- **Operations Manager** | CEDT Innovation Summit 2025 *(2025)*
- **Welfare Team Lead** | CEDT Ovenbreak 2025 *(2025)*
- **MC** | CEDT Guild Building 2025 *(2025)*
- **External Relations Team** | CEDT Innovation Summit 2026 *(2026)*
- **Sustain Team** | ESC Chulalongkorn University *(2026)*
- **Planning Team Member** | Vishnu 23rd Event *(2025)*
- **Registration Team Member** | ICPC Bangkok Regional 2025 Programming Contest *(2025)*
- **Registration Team Member** | CEDT Mock Exam *(2025)*

VOLUNTEER WORK

Graphic Designer | Lifefighter BKK Community, Bangkok *2022*

- Designed and developed a volunteer management system to streamline event coordination.